THE KINETICS OF HYDROGEN PEROXIDE DECOMPOSITION

(PART II)

OBJECTIVES

- To examine the effect of temperature on the rate of reaction and estimate the activation energy, E_a.
- To examine the effect of a catalyst on the rate of reaction.

INTRODUCTION

Studying the kinetics of a reaction is the key to understanding in detail how the reaction actually occurs—in other words, the *mechanism* of the reaction. In a kinetic study, one investigates the rates at which reactions occur and how those rates are affected by changes in the temperature, pressure, concentration, and the presence of catalysts. Once these factors are understood, conditions may be optimized to speed up or slow down a reaction as desired. In addition, information about reaction intermediates and electron movement during the reaction can be gained. A discussion of reaction rates, rate constants, laws and the theory of chemical kinetics may be found in your textbook.

The speed of a reaction is of practical interest, particularly in a commercial production setting. The speed—or rate—is usually described in the following way:

$$rate = \frac{change\ in\ concentration}{elapsed\ time} = \frac{\Delta[\quad]}{\Delta t}$$

where [ ] signifies the concentration in molarity ($moles/L$) of the compound you are observing. In this experiment, the quantity of O_2 gas produced will be monitored (instead of molarity) as a function of the elapsed time. The rate of reaction can then be written.

$$rate = \frac{\Delta(moles\ of\ O_2)}{\Delta t} = \frac{\Delta(O_2\ pressure)}{\Delta t}$$

(Recall $PV = nRT$ where the volume of the reaction chamber and T are held constant. Rate units are atm/min or $mm\ Hg/min$, depending on what pressure units are used.)

In this experiment you will investigate the kinetics of the reaction:

$$2\ H_2O_{2\,(g)} \rightarrow O_{2\,(g)} + 2\ H_2O_{(l)} \qquad (1)$$

Hydrogen peroxide, which contains oxygen in the oxidation state of -1, can act as both an oxidizing agent and as a reducing agent. H_2O_2 is unstable and disproportionates (meaning: can be oxidized and reduced in the same reaction) according to the half reactions:

$$H_2O_{2\,(aq)} + H_2O_{(l)} \rightarrow \frac{3}{2}O_{2\,(g)} + 4\,H^+ + 4\,e^- \quad (2)$$

$$H_2O_{2\,(aq)} + 2\,H^+ + 2\,e^- \rightarrow 2\,H_2O_{(l)} \quad (3)$$

While the conversion of H_2O_2 to O_2 and H_2O is normally slow, adding iodide as a catalyst will increase the rate of reaction. In a reaction equation, catalysts are shown above the arrow:

$$2\,H_2O_{2\,(g)} \overset{I^-}{\rightarrow} 2\,H_2O_{(l)} + O_{2\,(g)} \quad (4)$$

In this case, the I^- ion is considered a *homogeneous catalyst* because it is dissolved in the solution. While I^- affects the speed of the reaction and is involved in the rate-determining (or slow) step of the reaction, it is not part of the overall balanced equation because it is regenerated in the last step.

Activation Energy

In Week 2, we will explore how the rate constant (k) varies with temperature. Remember that when we report k values for a reaction, we report them at defined temperatures.

You will repeat the rate determination measurements from Week 1 for **Mixture 1 ONLY** but now at three different temperatures:

- Mixture 1 at reduced temperature (cold)
- Mixture 1 at room temperature (RT)
- Mixture 1 at elevated temperature (warm)

REMEMBER to note the initial (T_i) and final (T_f) reaction temperature of the mixture in each run. Make every effort to stabilize and accurately record these temperatures, as the reaction rates are sensitive to temperature fluctuations.

The relationship between k and temperature is given by the Arrhenius equation:

$$k = Ae^{-\frac{E_a}{RT}} \quad (6)$$

where R is the gas law constant ($8.314\frac{J}{mol \times K}$), T is temperature in Kelvin, E_a is the activation energy of the reaction, and A is the Arrhenius proportionality factor (unique to each reaction and sometimes referred to as the pre-exponential factor). Remember, e is the base of the natural logarithm and is equal to 2.71828. The activation energy is the minimum amount of energy required for a collision between reactants to result in products. Taking the natural logarithm of both sides provides a linear, more useful form of the Arrhenius equation:

$$\ln(k) = \ln(Ae^{-\frac{E_a}{RT}})$$

$$\ln(k) = \ln(A) + \ln(e^{-\frac{E_a}{RT}})$$

$$\ln(k) = \ln(A) + (-\frac{E_a}{RT})$$

Rearrange to the line form ($y = mx + b$):

$$\ln(k) = \frac{-E_a}{R} \times \left(\frac{1}{T}\right) + ln\ (A) \quad (7)$$

Thus, a plot of $\ln(k)$ (or any measure of $\ln(k)$ on the y-axis) versus $\frac{1}{T}$on the x-axis gives a straight line. The slope is

$$slope = \frac{-E_a}{R} \quad (8)$$

again using $R = 8.314 \frac{J}{mol \times K}$ to give $E_a = \frac{J}{mol}$. Usually, E_a is expressed in $\frac{kJ}{mol}$.

Table 1: Solution Ingredients for Mixture 1

Reagent	**Volume**
$0.10\ M\ KI$	$15.0\ mL$
$Distilled\ water$	$35.0\ mL$
$3\%\ H_2O_2\ (0.88\ M)$	$10.0\ mL$
Final volume	$60.0\ mL$

PROCEDURE

Part A. Temperature effects on the rate constant (Week 2)

1. Using the same apparatus set up as Week 1 (example in Figure 1), set up your glassware, stir bar and connect the probe.

2. BEGIN WITH THE COLD MIXTURE. Using a beaker, mix ice and water to create an ice/water bath with a temperature that measures 10℃ below room temperature. Record temperature in Table 2.

3. Mix the KI and water into the Erlenmeyer flask and place the Erlenmeyer flask in the ice/water bath. Wait as least five minutes for the mixture to reach the temperature (use the thermometer) of the bath.

4. Open Logger Pro and check the time to allow the reaction to run for at least 10 minutes. Check the sampling rate.

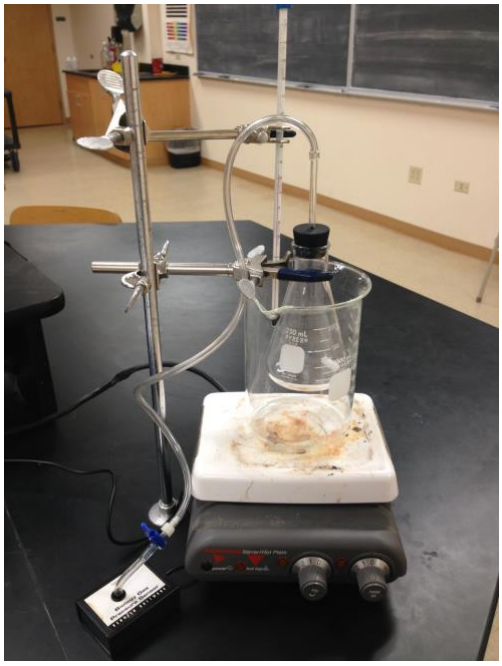


Figure 1. Apparatus set-up.

5. When all is ready, click COLLECT and quickly add the H_2O_2 to the reaction flask as completely as you can.

6. Insert the stopper firmly so it is gas tight and turn on the stir to a moderate rate. Maintain this stir rate throughout each experimental run.

7. Continue to collect data for at least 10 minutes. Select STOP to terminate data collection.

8. To analyze your data, select EXAMINE under the ANALYZE menu. Click and drag over the area of interest in your data - the most linear portion - to highlight it. This area will be 2-4 minutes long. Then choose Linear Fit under the ANALYZE menu. The equation for your selected data will appear on the data screen. Record this equation in your lab notebook, paying particular attention to the slope (rate), and round slope values to the appropriate significant figures (think about this).

9. Dispose of products in the waste container.

10. Repeat the data collection and analysis for the room temperature mixture, and then the elevated temperature mixture (10℃ above RT). Select COLLECT to begin logging the next data. When you select COLLECT a dialogue box will appear. Select STORE LATEST RUN.

CALCULATIONS

1. Calculate k for each experiment (i.e. each temperature) you ran today (Use Part 1 orders of reaction determined and Part 2 concentrations for Mixture 1). Fill in the following data table to help you plot the data.

Table 2: Data table for Arrhenius plot where temperature, T, is in Kelvins (K). The rate constant *k*, you will need to take the ln (*k*).

Trial	$\boldsymbol{k}$	$\boldsymbol{T}$ (℃)	$\boldsymbol{T}$ $\boldsymbol{(K)}$	$\boldsymbol{\frac{1}{T}}$	$\boldsymbol{ln(k)}$
Cool					
Room Temp					
Warm					

2. Using the following form of the Arrhenius equation: $\ln(k) = \frac{-E_a}{R} \times \left(\frac{1}{T}\right) + ln\ (A)$, to **create a scatter plot of your data** following the $y = mx + b$ form of the equation using Excel or a similar graphing program.

NOTE: These programs are powerful! You should be able to enter raw data such as temperature in degrees C (℃) and set up an equation in the program to convert all of your data into temperature in K. Please consult your instructor if you need any help with this!

3. Include a linear fit to your data, display an equation for the linear line fit and include all proper items for a good scatterplot (Title, labeled axes, appropriate units, etc.)
4. Your lab submission for this week should include Table 2 as well as your completed scatter plot.
5. Calculate and report the activation energy E_a (with proper units!).

INCLUDE THE FOLLOWING IN YOUR CONCLUSION

1. Write a conclusion that reports the observed E_a for the decomposition of H_2O_2.
2. Include in your conclusion thoughts you have on what this activation energy value tells us about the decomposition of H_2O_2.